

Lab overview

In this lab, I'll get hands-on experience testing my Python skills by updating application code for AWS Lambda functions that read and write directly from and to an Amazon DynamoDB table. The table data is added and retrieved through an Amazon Simple Storage Service (Amazon S3) hosted website that has an Amazon API Gateway integration to access the Lambda functions. The Lambda functions perform two of the four create, read, update, delete (CRUD) operations.

The website contains a button that puts items into the DynamoDB table and lists the current customers in the *LabCustomers* table. The Lambda function and code is already supplied for this feature. The website also features a form where I can enter new customer details, including the *customer ID*, *firstname*, and *lastname*. This form is connected to the *SaveCustomer* Lambda function, which currently doesn't perform any actions.

My task is to add functionality to this Lambda function. To do this, I need prior experience writing Python code to *PUT* data into an Amazon DynamoDB table, which I learned in the *Integrating Amazon DynamoDB (SPL-CX-100-CEIADD-1)* lab. I am challenged to update the Lambda function to retrieve the *ID*, *Firstname*, and *Lastname* from the Lambda *event* that's based on the submitted form from the Amazon S3 hosted website.

Next, I'll write the *put_item* code to save the item into the *LabCustomers* table. After I write the code, the last step is to use the AWS Serverless Application Model (AWS SAM) to deploy the updated code for the application.

Objectives

By the end of this lab, I am able to do the following:

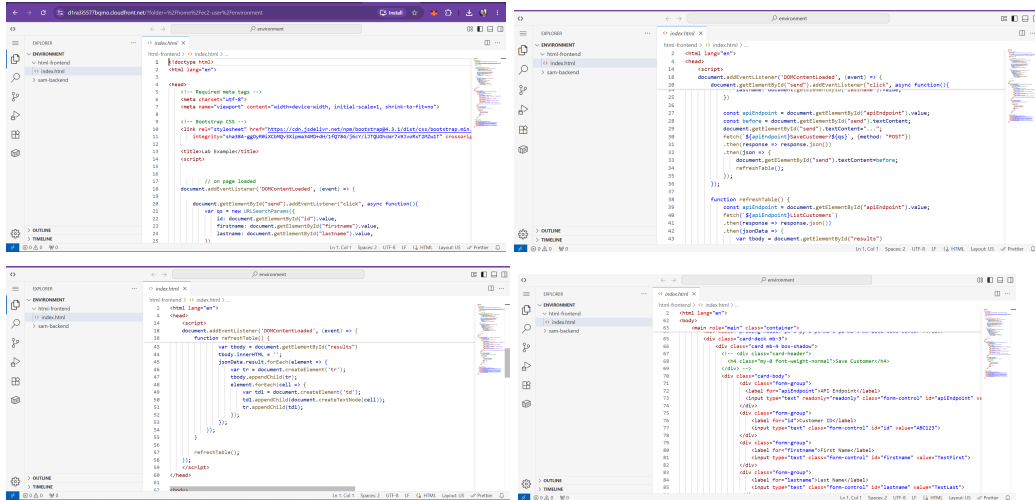
- Understand the AWS SDK for Python (Boto3) client code for DynamoDB in the application files.
- Update a Python application to PUT customer details into a DynamoDB table.
- Use Python code to update a DynamoDB table with parameters that are retrieved from an API Gateway request.

Task 1: Connect to the Code Editor IDE

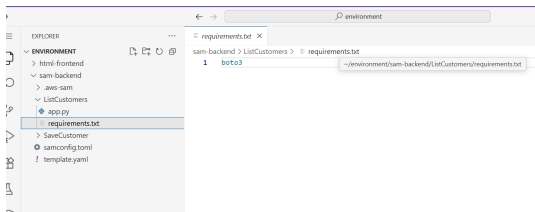
I'll now connect to the Code Editor IDE that's provisioned as part of this lab. Code Editor is the IDE that I'll use to review and update the Python code in this challenge.

Task 1.2: Review the application files

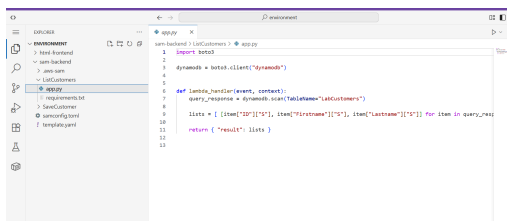
I'll expand the **html-frontend** folder and then open the **index.html** file. This HTML document is primarily used for interacting with an API endpoint to *save* and *list* customer data to and from a DynamoDB table that's named *LabCustomers*.



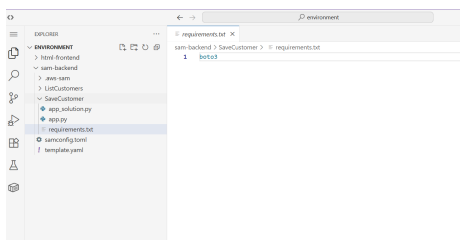
The **sam-backend** folder reveals two more folders named *ListCustomers* and *SaveCustomer*. The **ListCustomers** folder contains files named *app.py* and *requirements.txt*. The **ListCustomers/requirements.txt** file contains one Python package that's named *boto3*. This package is used by the Python application code.



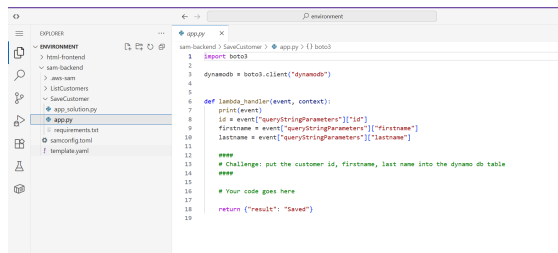
The **ListCustomers/app.py** reveals a python script that uses the SDK for Python to interact with the DynamoDB table that's named *LabCustomers*. In short, this script retrieves all items from the *LabCustomers* table in DynamoDB and returns them in a specific format.



The **SaveCustomer** folder has files, including *app.py* and *requirements.txt*. The *requirements.txt* file has a single dependency on the *boto3* package.



The `SaveCustomer/app.py` file is a Lambda function that's written in Python. It interacts with the DynamoDB service.

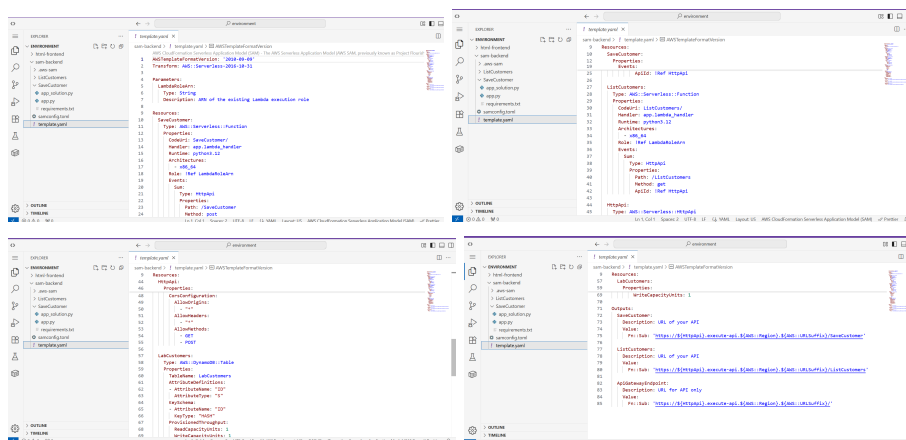


```
1 import boto3
2
3 dynamodb = boto3.client('dynamodb')
4
5
6 def lambda_handler(event, context):
7     print(event)
8     id = event['queryStringParameters']['id']
9     firstname = event['queryStringParameters']['firstname']
10    lastname = event['queryStringParameters']['lastname']
11
12    ###
13    # Challenge: put the customer id, firstname, last name into the dynamo db table
14    ###
15    # Your code goes here
16
17    return {"result": "Saved"}
```

The following list provides an overview of what the script does and what I need to update:

- **Imports the boto3 module:** Boto3 is the AWS SDK for Python. Python developers use the SDK for Python to write software that uses services such as Amazon Simple Storage Service (Amazon S3), Amazon Elastic Compute Cloud (Amazon EC2), and others.
- **Creates a DynamoDB client:** This action is done by using the `boto3.client` function and passing “dynamodb” as a parameter.
- **Defines a function that’s called `lambda_handler`, with `event` and `context` parameters:** This function is invoked when the Lambda function is called. The `event` parameter contains information about the event, and `context` contains runtime information.
- **Extracts `id`, `firstname`, and `lastname` from the `event` parameter:** The `event` parameter is expected to be a dictionary with a key that’s called `queryStringParameters`. In addition, `queryStringParameters` is itself a dictionary that contains keys for `id`, `firstname`, and `lastname`.
- **Inserts the extracted data into a DynamoDB table:** This section is the challenge portion of the lab that you need to complete.
- **Returns a dictionary with a key of `result` and a value of `Saved`:** This dictionary is the response from the Lambda function, and it indicates that the operation was successful.

The last file to review is the `template.yaml` file. It is a CloudFormation template that creates and updates two Lambda functions, a DynamoDB table, and an API Gateway HTTP API that the application uses. After you update the `SaveCustomer/app.py` file, this template is invoked by the AWS SAM CLI command to deploy the updated application files.

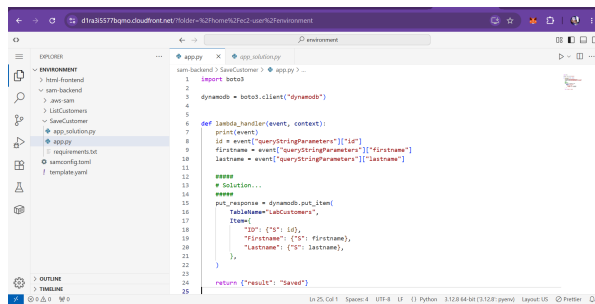


```
1 AWSTemplateFormatVersion: '2010-09-09'
2 Description: 'AWS Lambda Function: SaveCustomer'
3 Parameters:
4   - Name: SaveCustomerFunctionName
5     Type: String
6     Default: SaveCustomer
7   - Name: SaveCustomerFunctionRole
8     Type: String
9     Default: SaveCustomerRole
10  Resources:
11    SaveCustomerFunction:
12      Type: AWS::Lambda::Function
13      Properties:
14        Code: S3::Bucket::BucketName::BucketName
15        Handler: SaveCustomer.lambda_handler
16        Role: SaveCustomerRole
17        Runtime: python3.8
18        Architectures:
19          - x86_64
20    SaveCustomerRole:
21      Type: AWS::IAM::Role
22      Properties:
23        AssumeRolePolicyDocument:
24          Statement:
25            - Effect: Allow
26              Action: sts:AssumeRole
27              Principal:
28                AWS: iam::aws:iam::aws:role
29        Policies:
30          - PolicyName: SaveCustomerPolicy
31            PolicyDocument:
32              Statement:
33                - Effect: Allow
34                  Action: dynamodb:*
35                  Resource: '*'
36    SaveCustomerTable:
37      Type: AWS::DynamoDB::Table
38      Properties:
39        TableName: SaveCustomerTable
40        AttributeDefinitions:
41          - AttributeName: id
42            AttributeType: S
43          - AttributeName: firstname
44            AttributeType: S
45          - AttributeName: lastname
46            AttributeType: S
47        KeySchema:
48          - AttributeName: id
49            KeyType: HASH
50        ProvisionedThroughput:
51          ReadCapacityUnits: 1
52          WriteCapacityUnits: 1
53    SaveCustomerAPI:
54      Type: AWS::ApiGateway::HttpApi
55      Properties:
56        Name: SaveCustomerAPI
57        Description: SaveCustomerAPI
58        EndpointConfiguration:
59          EndpointType: REGIONAL
60        CorsConfiguration:
61          AllowedOrigins:
62            - '*'
63          AllowedMethods:
64            - GET
65            - POST
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```

Task complete: I have successfully reviewed the Amazon S3 hosted website that's used in this lab and observed its current lack of functionality. I have reviewed the files that create the overall application. I have identified the portion of the application code that I must update to add new items into the DynamoDB table. Lastly, I have reviewed the CloudFormation template that updates the application through the AWS SAM CLI.

Challenge A: Update the application

In this task, I am challenged with updating the *sam-backend/SaveCustomer/app.py* file with the missing code snippet to update the DynamoDB table with new customer details that are retrieved from the web form.

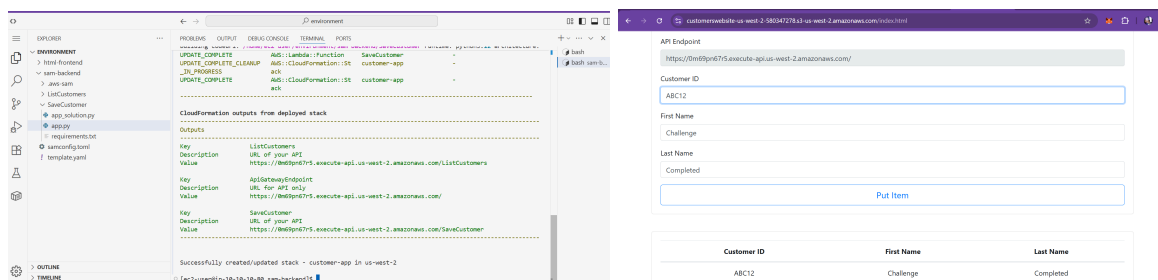


```
1 import boto3
2
3 dynamodb = boto3.client('dynamodb')
4
5
6 def lambda_handler(event, context):
7     print(event)
8     id = event['queryStringParameters']['id']
9     firstname = event['queryStringParameters']['firstname']
10    lastname = event['queryStringParameters']['lastname']
11
12
13    # Solution...
14    # Save
15    put_response = dynamodb.put_item(
16        TableName='LabCustomers',
17        Item={
18            'id': {'S': id},
19            'firstname': {'S': firstname},
20            'lastname': {'S': lastname},
21        },
22    )
23
24    return {"result": "Saved"}
```

Task complete: I have updated the *sam-backend/SaveCustomer/app.py* file with the missing code snippet. The code snippet updates the LabCustomers DynamoDB table with the new customer parameters that are retrieved from the web form by using API Gateway.

Task 2: Deploy the updates and test

In this task, I'll use AWS SAM to build and deploy the updated application package. After I deploy the updated application package, I then test the application. To test it, I'll return to the Amazon S3 hosted website and update the fields for the *customer ID*, *First Name*, and *Last Name*.



The left screenshot shows the AWS SAM CLI output in a terminal window. The output displays the CloudFormation outputs for the deployed stack, including the API Gateway endpoint URL. The right screenshot shows the Amazon S3 hosted website with the updated form fields: Customer ID (ABC12), First Name (Challenge), and Last Name (Completed). A 'Put Item' button is visible below the form.

Task complete: I have successfully deployed the latest application updates. I also tested the application to confirm that it stores new customer details in the DynamoDB table, and that it retrieves the table items to display in the Amazon S3 hosted website.

Conclusion

I have successfully done the following:

- Understood the AWS SDK for Python (Boto3) client code for DynamoDB in the application files.
- Updated a Python application to put customer details into a DynamoDB table.
- Used Python code to update a DynamoDB table with parameters that are retrieved from an API Gateway request.